

FFGFT — Information and Black Holes

*Topological Charge as Fundamental Information — Ontology vs.
Epistemology*

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This document clarifies what information means in FFGFT in the context of black holes. The **fundamental information** is not a quantity of bits, but a **topological charge** — comparable to a magnetic monopole charge, not to Shannon information. This charge is **topologically preserved**. The algebraic protection through $T \cdot m = 1$ (Dok. 078) and the interior operator at L_0 are fully described. The modified Hawking spectrum empirically grounds the access problem.

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1. The Hawking Information Paradox and Its Presuppositions

Hawking showed in 1974 that black holes emit thermal radiation. The paradox presupposes three conditions not all holding in FFGFT: **(1)** information = complete quantum state; **(2)** time is continuous; **(3)** singularity is physically real.

Why does the paradox arise at all? Hawking (implicitly) presupposes information is epistemic (what an observer can reconstruct) and time is continuous. Neither is fundamental in FFGFT: information is a topological charge, not an epistemic quantity; time is compactified on T^4 . The paradox dissolves by clarifying presuppositions, not by adding new physics.

2. What Information in FFGFT Is Not

2.1. Not our description — empirical grounding

Information in the FFGFT sense is not our knowledge of the black hole. The Hawking paradox is epistemic: can an external observer reconstruct the state? FFGFT answers: does the fundamental topological structure continue to exist?

Access problem, not existence problem — empirically grounded: If the topological information were truly destroyed, the Hawking radiation would be *exactly* black-body radiation. The **modified spectrum** due to the T^4 winding structure is the falsifiable difference. This makes the access problem physically tangible.

2.2. Not the sum of all detail states

Detail states can be epistemically lost through thermalisation — thermodynamic irreversibility, not ontological destruction.

3. What Information in FFGFT Is: Topological Charge

3.1. Topological invariants — no destruction operator

Definition of fundamental information: The topological charge (n_θ , n_ϕ , n_3 , n_4) on T^4 defines *what kind* of particles exist. Not a quantity of bits (Shannon), but a **topological charge** — comparable to a magnetic monopole charge or a winding number in string theory. This immediately decouples FFGFT from the bit-oriented information paradox debate.

No destruction operator acts on topological invariants. There is no physical process that can reduce winding numbers to zero without generating a measurable discontinuity. Classical singularities would be such a process — which is why they are structurally excluded in FFGFT. This statement belongs at the beginning, not the end.

3.2. Foundational relation as structure preservation

$$\tilde{T} \cdot m = 1 \Rightarrow \tilde{T} \rightarrow L_0/c = \xi \cdot t_P > 0$$

T-field never reaches zero. Winding structure is preserved.

3.3. Algebraic protection against singularities (Dok. 078)

From $\tilde{T} \cdot m = 1$: $\tilde{T} = 1/m$. If $m \rightarrow \infty$ then $\tilde{T} \rightarrow 0$ — asymptotically only. $\tilde{T} = 0$ exactly would require $m = \infty$ exactly — algebraically impossible.

Algebraic Protection — Dok. 078: • $m \rightarrow \infty$ implies $\tilde{T} \rightarrow 0$ only **asymptotically**, never exactly. • The singularity is an unreachable limit, not a forbidden state. • Not geometric, not quantum mechanical — **algebraic** from the foundational relation alone.

3.4. The interior operator: what happens at L_0

T-field at its minimum

$\tilde{T}_{\min} = \xi \cdot t_P > 0$ — finite, not singular.

Maximum winding density

$1/L_0^4$ per spacetime volume — analogous to the lower bound in a lattice with finite volume. Level $N = 0$ of the ξ ladder. Below this the state space ends.

Lagrangian stiffness (Dok. 201)

$-\frac{1}{2} m_{\tilde{T}}^2 (\partial m)^2$ with $m_{\tilde{T}} = \xi/\ell$ prevents divergence of ∂m . Algebraic property of the field equations.

Topological bounce condition

No states below $N = 0 \Rightarrow$ state space boundary, not a force-counterforce mechanism.

Formal criterion for the transition operator

An interior operator would need to describe the mapping $(n_i) \rightarrow$ extremal configuration at L_0 — under preservation of topological charge, with maximum winding density. The T^4 algebra suggests **unitarity**, but non-trivial commutation with the thermodynamic limit. This is the open research question — precisely stated, not a fundamental gap.

Interior of a black hole in FFGFT — not unknown territory: • $\tilde{T}_{\min} = \xi \cdot t_P > 0$ • Maximum winding density: $1/L_0^4$ • Lagrangian stiffness: ∂m does not diverge • State space boundary at $N = 0$ • Unitary transition operator (details open)

3.5. Fundamental information is not the total description

Fundamental information = **class** (topological charge), not **instance** (quantum state). FFGFT preserves the class.

4. Black Holes as Inversion — Not as Destruction

Black hole = inversion of normal matter structure: maximum compression, T-field at minimum. What fell in as an electron remains topologically classifiable as an electron.

Two levels — do not confuse: Thermodynamic: directed, irreversible, entropy increases. Topological: winding numbers **topologically preserved**. No destruction operator acts on topological invariants. The Hawking paradox arises from confusing these levels.

5. The Compactified Time Dimension

Compactification means periodicity of field configurations in the time direction — not reversibility of thermodynamic time. A macroscopic process remains irreversible. Our experienced time direction is thermodynamically emergent, not fundamental. The T^4 topology is time-direction-neutral — this does not mystically overload the theory.

"Forever lost in the past" presupposes an absolute ontological past boundary. In a compactified time dimension, this absoluteness is not self-evident. n_4 remains topologically preserved.

6. What FFGFT Claims — and What It Does Not

Claimed: (1) No singularity: $L_0 = \xi \cdot l_P$ as lower bound. (2) Topological charge preserved: algebraic protection through $T \cdot m = 1$. (3) Modified Hawking spectrum: deviation from black-body (falsifiable). (4) Gravitational wave echoes after mergers.

Not claimed: Complete state reconstruction · Carrier recovery · Fully specified transition operator near $N=0$ · Suspension of thermodynamic irreversibility.

7. Connection to the ξ Recursion

The ξ ladder defines stable configurations. Black holes: extreme states at the lower end ($N = 0$) — within the ξ structure, not outside it. Topologically well-determined.

Summary

Question	FFGFT Answer
Why does the paradox arise?	Hawking presupposes information as epistemic and time as continuous — neither fundamental in FFGFT
What is fundamental information?	Topological charge: $(n_\theta, n_\phi, n_3, n_4)$ on T^4 — not bits

Question	FFGFT Answer
Is information lost?	Topological charge: no (topologically preserved). Detail state: access problem
Empirical grounding	Modified Hawking spectrum: If information were destroyed, radiation would be exactly thermal
Algebraic protection (Dok. 078)	$m \rightarrow \infty$ implies $T \rightarrow 0$ only asymptotically — singularity unreachable
What happens at L_0 ?	$T_{\min} > 0 \cdot \text{Max. winding density} \cdot \text{Lagrangian stiffness} \cdot \text{State space boundary } N=0$
Is there a singularity?	No — L_0 as boundary; state space ends at $N=0$
Irreversible?	Thermodynamically yes. Topologically: fundamental structure remains
Time compactification	Periodicity of field configurations, not reversal of thermodynamic time
Falsifiable tests	Hawking spectrum (not exactly thermal); gravitational wave echoes
What remains open?	Transition operator $(n_i) \rightarrow \text{extremal configuration at } L_0$ unitary, details open

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